School of Computer Science and Engineering (SCOPE)

BCSE302P – Database Systems Lab

Slot: D1 (L51 + L52) · Complete Project Report

Project Title

Policy Intelligence for Agriculture: NLP and LLM-Based Analysis of Farmer Feedback on Government Welfare Schemes

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1. Structured Abstract

Table 1: Structured abstract

Component	Summary
Background	Government welfare schemes for farmers are evaluated mainly through satisfaction ratings, although farmers also describe their experiences in open-ended answers that are rarely analysed.
Objective	To identify the problems farmers face and the changes they request, to test whether satisfaction scores can be taken at face value, and to determine which NLP approach interprets farmer feedback reliably.
Data	Field survey of 1,500 farmers in Chittoor district, Andhra Pradesh, verified by the Ethnic Community, Government of Andhra Pradesh; 1,214 farmers provided written feedback.
Methods	Lexicon and transformer sentiment analysis, aspect-based sentiment analysis, topic modelling, suggestion mining and rating–text triangulation; comparison of 11 sentiment models, including three large language models, on 243 human-labelled test responses.
Results	73.7% of farmers know at least one scheme, but awareness converts unevenly into enrolment (89.4% for Rythu Bharosa, 9.1% for Crop Insurance). Ratings and written feedback disagree in 28.7% of timeliness and 28.3% of enrolment cases ($p < 0.001$). The best model, Fine-tuned ModernBERT-base, achieved a macro-F1 of 0.765; the VADER lexicon achieved 0.223.
Conclusion	Satisfaction scores alone conceal enrolment and timing problems, and general-purpose sentiment tools are unreliable on farmer feedback. Validated NLP models make written feedback usable as policy evidence. The work is being prepared as a research paper targeting a Q1 journal.

2. Introduction

Agriculture employs nearly half of India's workforce, and central and state governments operate numerous welfare schemes, including PM-KISAN, Rythu Bharosa, crop insurance, minimum support price mechanisms and input subsidy programmes, to improve the income security and resilience of farming households. Despite substantial public investment, scheme impact remains uneven: limited awareness, delayed disbursement and administrative bottlenecks are repeatedly reported as barriers.



Figure 1: Smallholder farming in Andhra Pradesh

Source: Wikimedia Commons · PJeganathan · CC BY-SA 3.0



Figure 2: Farmland at Gudimallam, Chittoor district

Source: Wikimedia Commons · Ms Sarah Welch · CC0

In Andhra Pradesh, many schemes are delivered through village-level service centres known as Rythu Bharosa Kendrams, where farmers register, obtain information and receive support. The success of these schemes is conventionally measured by asking farmers to rate their satisfaction on a fixed scale. Such ratings are convenient to collect and summarise, but they compress lived experience into a narrow numeric range and are susceptible to acquiescence and social-desirability bias.



Figure 3: A Rythu Bharosa Kendram, the village-level service centre for farm schemes

Source: Wikimedia Commons · Veera Ram Kumar · CC0

Natural Language Processing (NLP) offers a scalable way to extract sentiment and recurring themes directly from farmers' own words, and to compare these with the ratings given by the same respondents. A disagreement between what a farmer rates and what the same farmer writes is itself informative: it reveals service problems that fixed-choice questions fail to capture.

3. Problem Statement

Welfare schemes can only improve farmers' lives if farmers know about them, are able to enrol, and receive benefits on time. Evidence from the surveyed farmers indicates failures at each of these stages:

Table 2: Problems identified among the surveyed farmers

Problem	Evidence from the survey
Many farmers are not aware of any scheme	387 of 1,469 respondents (26.3%) were not aware of any welfare scheme.
Awareness does not lead to utilisation	On average, a farmer had heard of 2.41 schemes but was enrolled in only 1.61; 51.5% of farmers knew more schemes than they used.
Utilisation differs sharply between schemes	Enrolment among aware farmers ranged from 89.4% (Rythu Bharosa) to 2.2% (PM Krishi Sinchai Yojana).
Benefits arrive late and procedures are complex	Payment delays and late credit account for about 44% of complaint text; registration, documentation and office approval form most of the remainder.
Evaluation overlooks farmers' own words	1,214 farmers described their experience in writing, yet schemes are judged by ratings that may not reflect these accounts.

A further practical obstacle is that the written answers are unstructured, inconsistently spelt and recorded in English after interviews conducted mainly in Telugu. General-purpose sentiment tools have not been validated on such text, so their output cannot be assumed to be correct.

Research question: What do farmers' own words reveal about the awareness, utilisation and delivery of welfare schemes, how far do satisfaction ratings reflect this experience, and which NLP approach interprets farmer feedback reliably enough to inform policy?

4. Research Gap

- Research on farmers' awareness of Indian welfare schemes relies almost entirely on structured survey percentages, with little systematic comparison against the respondents' own written accounts from the same interview.
- NLP and aspect-based sentiment analysis have proven effective on open-ended survey text in human resources, healthcare and policy discourse, but remain largely unapplied to the evaluation of agricultural welfare schemes with rural, often low-literacy respondents.
- No identified study triangulates NLP-derived sentiment against structured satisfaction ratings from the same farmers, disaggregates the comparison by scheme and service aspect, or compares complaint themes with independently mined farmer suggestions.
- Few studies validate sentiment tools against human labels before reporting their output, and none was found that compares domain-specific agricultural LLMs with general models on farmer feedback.

5. Objectives and Scope

5.1 Objectives

1. **Identify the problems farmers actually face**, in their own words rather than in pre-set survey categories.
2. **Identify what farmers are asking for**, mined directly from their improvement suggestions.
3. **Test whether structured satisfaction scores can be taken at face value**, and identify which schemes and service dimensions underperform.
4. **Measure the gap between awareness and utilisation** of each scheme.
5. **Determine which NLP approach interprets farmer feedback reliably**, by comparing lexicon, transformer and large-language-model methods against human-labelled responses.
6. **Translate each finding into a specific recommendation** for the Department of Agriculture and Farmers Welfare, supported by the underlying evidence.

5.2 Scope

Table 3: Project scope

Within scope	Outside scope
1,500 interviews from Chittoor district, Andhra Pradesh	Other districts and states
Sentiment, aspect-based sentiment, topics and suggestions in written feedback	Collection of new survey data
Comparison of eleven sentiment models, including LLMs	Training new large language models from scratch
Statistical testing of all findings	Causal claims about scheme impact
Policy recommendations and a research paper draft	Publication of farmers' personal information

6. Dataset Description

Data were collected from 1,500 farmers within a 50 km radius of the university campus in Chittoor district, Andhra Pradesh, through structured interviews conducted by trained investigators. The dataset was verified by the Ethnic Community, Government of Andhra Pradesh. Interviews were conducted mainly in Telugu (90.5%), and answers were recorded in English.

Table 4: Structure of the survey instrument

Section	Questions	Content
A — Profile	12	age, gender, village, education, landholding, crops, income, experience
B — Awareness	5	schemes heard of, schemes enrolled in, information sources
C — Feedback	11	satisfaction, ratings of six service aspects, three open-ended answers (positive aspects, problems, suggestions)
D — Outlook	4	rating of government support, trend, confidence
E — Interview	4	language, comfort, observations, duration

Table 5: Profile of the respondents

Characteristic	Value
Respondents	1,500
Respondents with written feedback	1,214
Mean age	50.7 years
Mean farming experience	19.8 years
Median landholding	2.05 acres
Gender	67.2% male, 32.8% female
Marginal farmers (< 2.5 acres)	53.3%
Principal crops	mango, paddy, sugarcane
Aware of at least one scheme	73.7%

7. Tools and Technologies Used

Table 6: Tools and technologies

Category	Technology	Purpose
Programming language	Python 3.12	complete analysis pipeline
Data handling	pandas, openpyxl, RapidFuzz	reading survey files, cleaning, matching misspelt scheme names
Lexicon NLP	NLTK VADER with an agricultural word list	rule-based sentiment scoring
Classical machine learning	scikit-learn (TF-IDF, logistic regression, LDA)	trained baseline and topic modelling
Transformer NLP	Hugging Face Transformers, SetFit	RoBERTa, DeBERTa-v3, ModernBERT and few-label learning
Large language models	Dhenu2 (KissanAI), Qwen2.5-3B, Claude	classification of feedback using labelling guidelines
GPU computation	PyTorch (CUDA), bitsandbytes 4-bit quantisation, NVIDIA RTX 2050	execution of 3-billion-parameter models on a 4 GB GPU
Statistics	SciPy	chi-square, McNemar and McNemar–Bowker tests, bootstrap confidence intervals
Data storage	PostgreSQL 16	storage of survey responses, model outputs and human labels
Reporting	python-docx, Power BI dataset	statistical report, research paper draft, dashboards
Development tools	VS Code, Git and GitHub, Google Colab	development, version control, optional cloud GPU

8. Project Folder Structure

The project is organised so that every result can be regenerated from the original survey file. Figure 4 shows the folder structure, and the tables that follow explain each part.

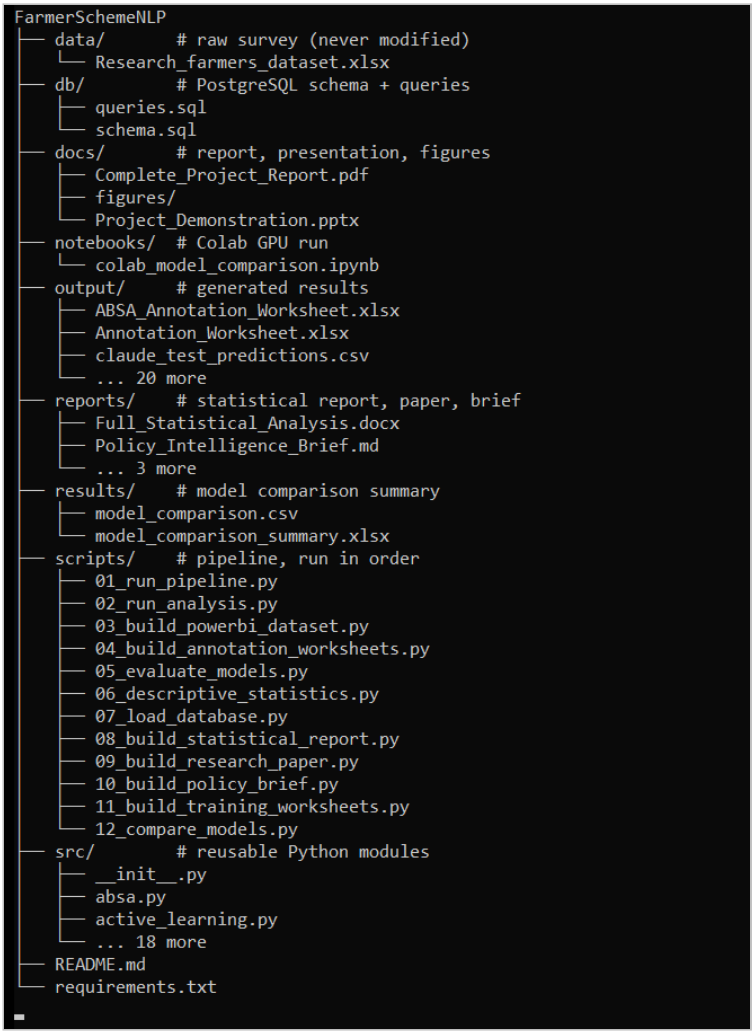


Figure 4: Project folder structure

Table 7: Top-level folders

Folder / file	Description
data/	the original survey workbook (Research_farmers_dataset.xlsx); never modified, and excluded from the public repository
src/	reusable Python modules containing the analysis logic (described in Table 8)
scripts/	twelve numbered scripts that run the analysis in order (described in Table 9)
notebooks/	Google Colab notebook for running the large language models on a cloud GPU
output/	all generated results: master sentiment file, statistics workbook, topic results, labelling worksheets, model predictions
results/	public summary of the model comparison (no personal data)
docs/	final deliverables: this report, the presentation and all figures
reports/	statistical report, research paper draft and policy brief
db/	storage schema for the survey responses and model outputs
README.md, requirements.txt	project description, execution steps and pinned library versions

Table 8: Analysis modules

Module (src/)	Function
preprocess.py, preprocess_nlp.py	text cleaning, normalisation and removal of non-informative answers such as "No" or "Nil"
schemes.py	mapping of 93 free-text scheme spellings to 11 canonical schemes using fuzzy matching
sentiment_lexicon.py	VADER sentiment with an agricultural domain word list
sentiment_transformer.py	RoBERTa transformer sentiment
absa.py	aspect-based sentiment for the six service aspects
topic_model.py	LDA topic modelling with model-selection diagnostics
suggestions.py	mining of improvement suggestions
triangulate.py	comparison of written sentiment with the farmer's own ratings
statistical_tests.py, crosstabs.py, descriptive.py	chi-square, correlation and McNemar–Bowker tests; descriptive tables
data_quality.py	detection of contradictory or implausible answers
evaluation.py	accuracy, macro-F1, kappa and bootstrap confidence intervals
active_learning.py, worksheets.py	selection of training responses and creation of labelling worksheets
model_comparison.py	training, prompting and evaluation of all eleven models
report_docx.py, excel_style.py, config.py	report generation, formatting and shared settings

Table 9: Pipeline scripts

Script (scripts/)	Function
01_run_pipeline	cleaning, scheme matching, VADER and RoBERTa sentiment, aspect sentiment
02_run_analysis	data quality, statistical tests, topic models
03_build_powerbi_dataset	dashboard dataset
04_build_annotation_worksheets	stratified 20% test sample for human labelling
05_evaluate_models	accuracy of the pipeline models against human labels
06_descriptive_statistics	all descriptive tables
07_load_database	storage of responses, model outputs and labels, with 41 consistency checks
08 – 10	statistical report, research paper draft and policy brief
11_build_training_worksheets	400 training responses selected by uncertainty and at random
12_compare_models	complete comparison of eleven models

9. Methodology and Workflow

The workflow converts the raw survey into validated findings in five stages: ingestion and cleaning, NLP analysis, model validation, storage, and generation of outputs (Figure 5).

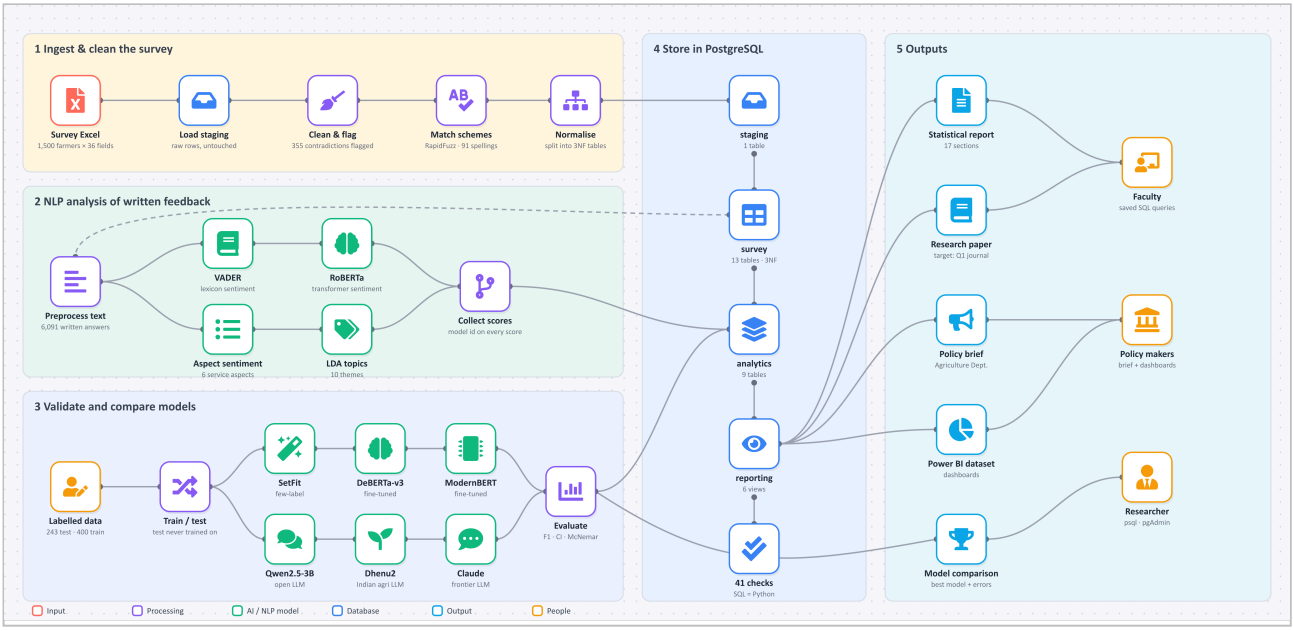


Figure 5: End-to-end project workflow

9.1 Text Preprocessing

The three open-ended answers of each farmer (positive aspects, problems faced and suggestions) were cleaned and normalised. Answers consisting only of non-informative phrases such as "No", "Nothing" or "No problems" were treated as empty rather than as sentiment, since lexicon tools otherwise score them as negative. Scheme names written in 93 different ways were mapped to 11 canonical schemes using fuzzy matching.

9.2 Analysis Methods

Table 10: Analysis methods

Method	Purpose	Output
Overall sentiment (VADER, RoBERTa)	polarity of each farmer's written feedback	Positive / Neutral / Negative per farmer
Aspect-based sentiment analysis	sentiment towards each of the six rated service aspects	2,659 aspect-level scores
Topic modelling (LDA)	recurring themes in complaints and in positive feedback	five complaint topics and five satisfaction topics
Suggestion mining	changes most frequently requested by farmers	ranked themes from 927 suggestions
Rating–text triangulation	agreement between a farmer's rating and written sentiment	divergence rate per aspect
Statistical testing	significance of differences and associations	chi-square, McNemar–Bowker, Spearman correlation

9.3 System Architecture

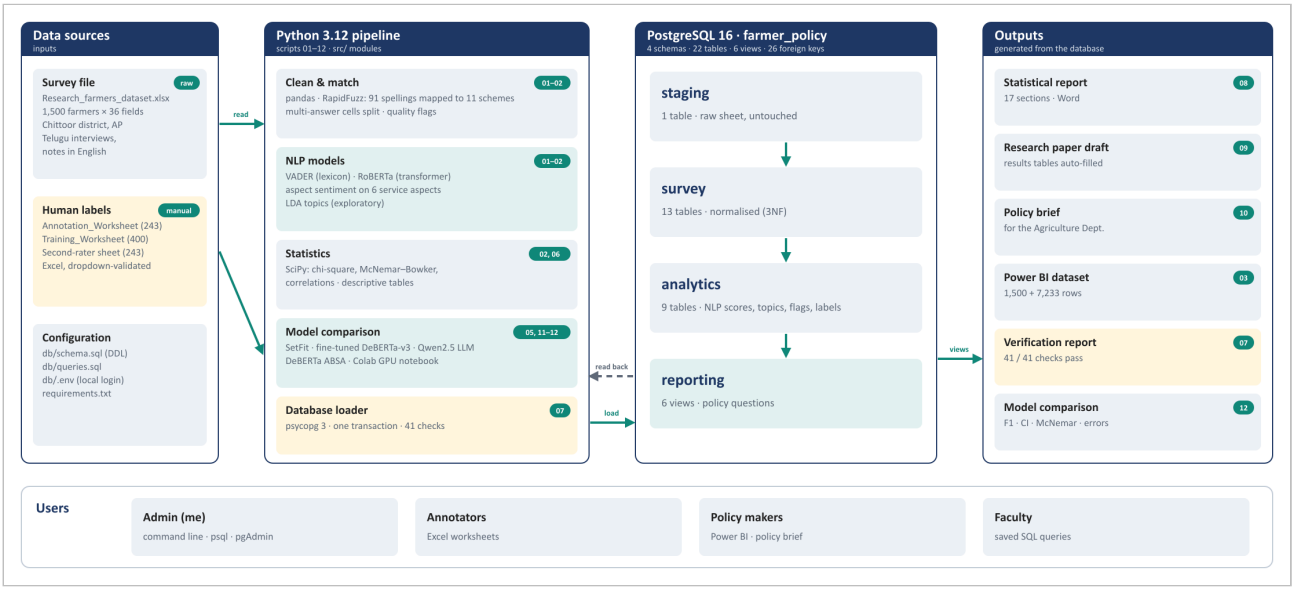


Figure 6: System architecture: inputs, analysis pipeline, storage and outputs

10. NLP Models and Large Language Models

10.1 Types of NLP Compared

Table 11: Types of NLP compared

NLP type	Model(s)	Method
Lexicon-based sentiment	VADER with agricultural lexicon	aggregates the polarity of known words
Pre-trained transformer	RoBERTa (trained on social-media text)	applied without further training
Classical machine learning	TF-IDF with logistic regression	trained on labelled farmer responses
Few-label learning	SetFit (all-mpnet-base-v2)	contrastive learning from a few hundred examples
Fine-tuning	DeBERTa-v3, ModernBERT	full training of a transformer on farmer responses
LLM prompting	Dhenu2, Qwen2.5-3B, Claude	classification guided by the labelling instructions, with zero or three examples
Aspect-based sentiment	keyword method; DeBERTa ABSA	sentiment towards each service aspect separately

10.2 Large Language Models

Table 12: Large language models and encoders

Model	Developer	Parameters	Reason for inclusion
Dhenu2 (Llama-3.2 based)	KissanAI, India	3 billion	domain-specific model for Indian agriculture
Qwen2.5-3B-Instruct	Alibaba	3 billion	general-purpose model of identical size, for a controlled comparison
Claude Opus 5.5	Anthropic	frontier-scale	state-of-the-art reference model
ModernBERT, DeBERTa-v3	Answer.AI, Microsoft	approx. 150 million	encoder models suitable for full fine-tuning

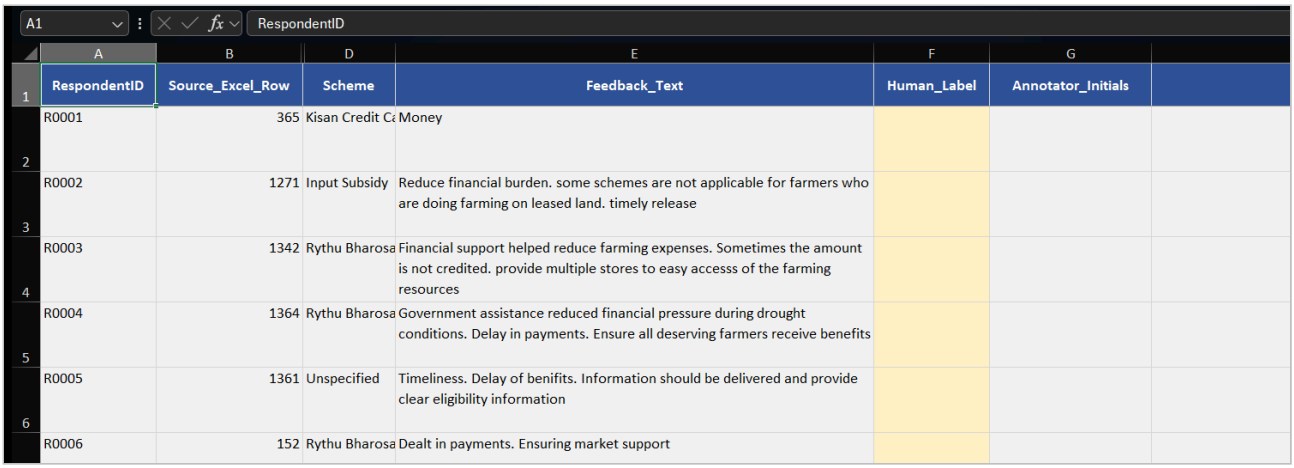
The 3-billion-parameter models were loaded with 4-bit quantisation so that they could run on a 4 GB laptop GPU. Each LLM was evaluated both zero-shot and with three labelled examples per class.

10.3 Evaluation Design

Table 13: Evaluation datasets

Dataset	Size	Purpose
Test set	243 responses	stratified random 20% sample; used only for evaluation
Training set	400 responses	training of trainable models and selection of few-shot examples
Aspect test set	131 responses (274 aspect mentions)	evaluation of aspect-level models

Labels were produced jointly by the author and one additional annotator using a written labelling guide. Training-set labels were first suggested by an AI model and then verified by the annotators, who changed 86 of 400 of the suggestions. Each trainable model was run with five random seeds. Results are reported with 95% bootstrap confidence intervals, and differences between models were tested with the exact McNemar test.



	A	B	D	E	F	G
1	RespondentID	Source_Excel_Row	Scheme	Feedback_Text	Human_Label	Annotator_Initials
2	R0001	365	Kisan Credit Ca	Money		
3	R0002	1271	Input Subsidy	Reduce financial burden. some schemes are not applicable for farmers who are doing farming on leased land. timely release		
4	R0003	1342	Rythu Bharosa	Financial support helped reduce farming expenses. Sometimes the amount is not credited. provide multiple stores to easy accesss of the farming resources		
5	R0004	1364	Rythu Bharosa	Government assistance reduced financial pressure during drought conditions. Delay in payments. Ensure all deserving farmers receive benefits		
6	R0005	1361	Unspecified	Timeliness. Delay of benifits. Information should be delivered and provide clear eligibility information		
	R0006	152	Rythu Bharosa	Dealt in payments. Ensuring market support		

Figure 7: Labelling worksheet used by the annotators (participant names hidden)

11. Result Analysis

11.1 Problems Farmers Face (Objective 1)

Topic modelling of the problem answers identified five complaint themes. Payment timing dominates: late credit and payment delays together account for about 44% of complaint text.



Figure 8: Share of complaint text by topic (%)

Table 14: Complaint themes identified by topic modelling

Complaint theme	Share	Representative terms
Payment delays & bank verification	21.7%	payment, delayed, bank, delay, eligible, family
Online registration & documents	18.3%	application, online, registration, document, required, delay
Insufficient support for crop needs	16.8%	need, insufficient, crop, official, support, requirement
Late credit of benefits	22.1%	time, delay, sufficient, receiving, late, credited
Low awareness & slow office approval	21.1%	time, application, lack, long, difficulty, awareness

Table 15: Sentiment towards each service aspect

Service aspect	Mentions	% Negative	% Neutral	% Positive
Timeliness	496	39.9	31.7	28.4
Amount of support	786	8.8	26.8	64.4
Enrollment ease	415	45.1	40.0	14.9
Official behavior	96	12.5	12.5	75.0
Clarity of information	178	30.3	30.9	38.8
Overall impact	688	7.0	20.8	72.2

Enrolment ease (45.1% negative) and timeliness (39.9% negative) are the most criticised aspects, while the amount of support and the overall impact are described mostly positively.

11.2 What Farmers Ask For (Objective 2)

Table 16: Improvement suggestions (927 substantive suggestions)

Rank	Suggestion theme	Representative terms (frequency)
1	Increase the benefit amount	"amount" (233), "increase" (212)
2	Simplify / speed up the application process	"process" (99), "application" (88), "reduce" (74), "procedures" (30)
3	Run more awareness programs	"awareness" (78), "programmes" (36), "programs" (32), "conduct" (37)
4	Ensure faster, more reliable disbursement	"timely" (33), "faster" (32), "ensure" (32)

Complaints are dominated by payment timing, whereas suggestions place a larger benefit amount first. Delays are the problem farmers experience; a larger amount is the change they most request.

11.3 Can Satisfaction Scores Be Taken at Face Value? (Objective 3)

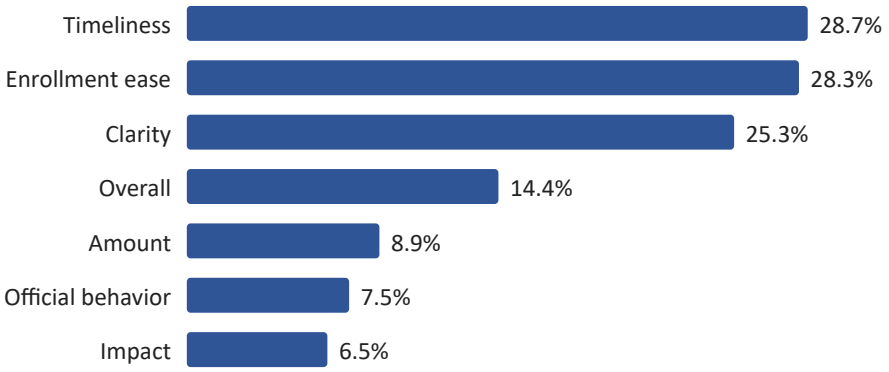


Figure 9: Disagreement between ratings and written sentiment (%)

Table 17: Rating–text divergence by aspect

Comparison	Comparable responses	Divergent	% divergent
Timeliness	484	139	28.7
Enrollment ease	399	113	28.3
Clarity	174	44	25.3
Overall	1197	172	14.4
Amount	760	68	8.9
Official behavior	93	7	7.5
Impact	666	43	6.5

A McNemar–Bowker test confirms that the disagreement between ratings and written sentiment is systematic ($\chi^2 = 192.73$, df = 3, n = 1,197, $p < 0.001$). The divergence is concentrated in timeliness and enrolment, the same aspects that receive the most negative written feedback.

Table 18: Scheme-level sentiment and ratings (schemes with n ≥ 30)

Scheme	n	% Positive	% Negative	Avg. satisfaction (−2 to +2)	Avg. impact (−2 to +2)
Rythu Bharosa	689	64.7	16.8	1.01	0.88
PM-KISAN	288	70.8	11.8	1.02	0.74
Kisan Credit Card	84	70.2	13.1	1.17	1.06
Input Subsidy	62	56.5	11.3	1.14	1.02
Crop Insurance	31	41.9	45.2	0.97	1.03

Table 19: Weakest service aspect of each major scheme

Scheme	Weakest aspect	% Negative mentions
Input Subsidy	Enrolment ease	53.8%
Kisan Credit Card	Enrolment ease	50.0%
PM-KISAN	Enrolment ease	47.6%
Rythu Bharosa	Enrolment ease	40.6%

Enrolment ease is the weakest aspect of all four major schemes, indicating a shared registration, documentation or verification bottleneck. Crop Insurance has the highest negative share (45.2%), and PM-KISAN combines high satisfaction with the lowest self-reported impact on farming.

11.4 Gap Between Awareness and Utilisation (Objective 4)

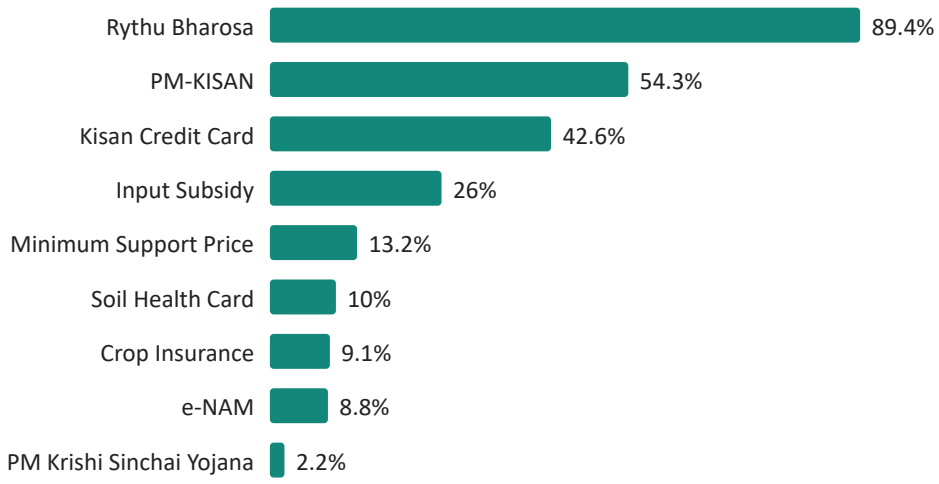


Figure 10: Enrolment among farmers aware of each scheme (%)

Table 20: Awareness-to-enrolment conversion by scheme

Scheme	Heard of	Heard and enrolled	Conversion
Rythu Bharosa	1104	987	89.4%
PM-KISAN	917	498	54.3%
Input Subsidy	388	101	26.0%
Crop Insurance	143	13	9.1%
Kisan Credit Card	136	58	42.6%
Minimum Support Price	106	14	13.2%
PM Krishi Sinchai Yojana	93	2	2.2%
Soil Health Card	50	5	10.0%
e-NAM	34	3	8.8%

73.7% of farmers are aware of at least one scheme, but farmers had heard of 2.41 schemes on average while being enrolled in only 1.61, and 51.5% knew of more schemes than they used. Schemes such as Crop Insurance (9.1%) and PM Krishi Sinchai Yojana (2.2%) are known but rarely joined, so the barrier lies in enrolment rather than awareness.

11.5 Differences Between Farmer Groups

Table 21: Chi-square tests of sentiment against farmer characteristics

Characteristic	χ^2	df	p-value	Cramér's V
Education	58.53	16	9.2e-07	0.143
Landholding	32.89	6	1.1e-05	0.125
Info Source	27.42	16	3.7e-02	0.107
Scheme	42.32	18	1.0e-03	0.133
Gender	20.37	2	3.8e-05	0.119
Income	48.2	10	5.7e-07	0.13

All associations are statistically significant but weak (Cramér's V 0.11–0.14). Income shows the clearest gradient: negative sentiment falls from 36.1% among the lowest-income farmers to 5.4% among the highest (Table 23).

Table 22: Sentiment by income group

Annual income	% Positive	% Neutral	% Negative	n
Below ₹25,000	45.9	18.0	36.1	122
₹25,000–50,000	47.0	24.7	28.3	279
₹50,000–1,00,000	56.5	21.1	22.4	522
₹2,00,000–5,00,000	45.4	33.0	21.6	97
₹1,00,000–2,00,000	62.9	19.7	17.4	380
Above ₹5,00,000	81.1	13.5	5.4	37

12. Model Comparison (Objective 5)

All eleven models were evaluated on the same 243 test responses, none of which were used for training. Macro-F1 is the primary metric because it weights the positive, neutral and negative classes equally and is not inflated by the dominant neutral class. The test set contained neutral 143, positive 76, negative 24 responses.

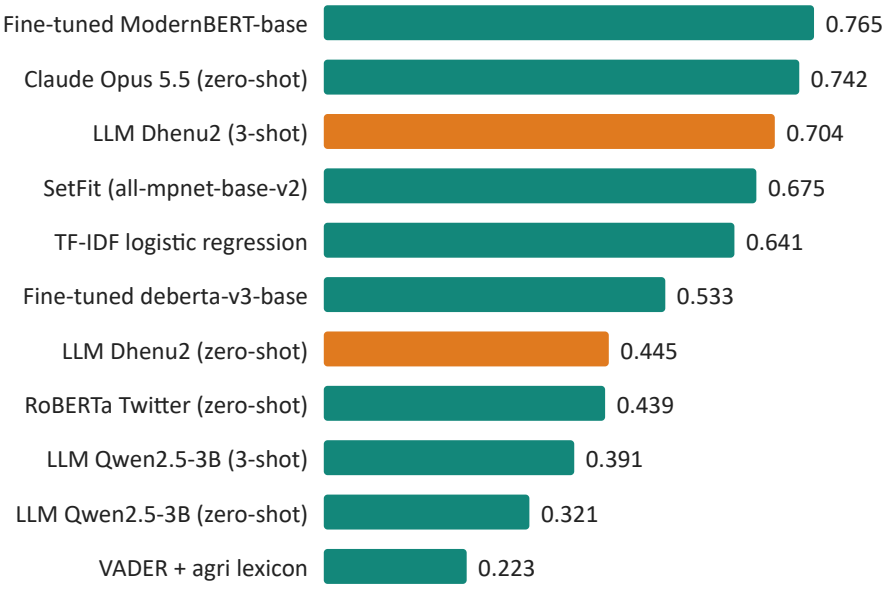


Figure 11: Macro-F1 of all evaluated models (Dhenu2 highlighted)

Table 23: Performance of all models on the test set

Model	Type	Accuracy	Macro-P	Macro-R	Macro-F1	95% CI (F1)	Cohen's κ	Mean ± SD (5 seeds)
Fine-tuned ModernBERT-base	trained (fine-tuned)	0.835	0.810	0.736	0.765	0.687 - 0.832	0.683	0.750 ± 0.034
Claude Opus 5.5 (zero-shot)	LLM (prompted, external)	0.786	0.746	0.787	0.742	0.676 - 0.801	0.620	—
LLM Dhenu2-In-Llama3.2-3B-Instruct (3-shot)	LLM (prompted)	0.749	0.682	0.746	0.704	0.63 - 0.766	0.566	—
SetFit (all-mpnet-base-v2)	trained (few-label)	0.840	0.909	0.647	0.675	0.592 - 0.752	0.675	0.685 ± 0.022
TF-IDF logistic regression	trained baseline	0.745	0.708	0.618	0.641	0.562 - 0.718	0.491	—
Fine-tuned deberta-v3-base	trained (fine-tuned)	0.749	0.838	0.534	0.533	0.476 - 0.595	0.481	0.584 ± 0.080
LLM Dhenu2-In-Llama3.2-3B-Instruct (zero-shot)	LLM (prompted)	0.519	0.445	0.478	0.445	0.38 - 0.51	0.208	—
RoBERTa Twitter (zero-shot)	zero-shot baseline	0.506	0.445	0.471	0.439	0.367 - 0.512	0.103	—
LLM Qwen2.5-3B-Instruct (3-shot)	LLM (prompted)	0.457	0.445	0.472	0.391	0.317 - 0.458	0.065	—
LLM Qwen2.5-3B-Instruct (zero-shot)	LLM (prompted)	0.300	0.350	0.511	0.321	0.268 - 0.376	0.048	—
VADER + agri lexicon	zero-shot baseline	0.235	0.215	0.296	0.223	0.163 - 0.279	-0.144	—

Best model: Fine-tuned ModernBERT-base (macro-F1 0.765, accuracy 83.5%). Its advantage over Claude Opus 5.5 (zero-shot), SetFit (all-mpnet-base-v2) is not statistically significant, so these models are regarded as jointly best; it is significantly better than all other models (exact McNemar test, $p < 0.05$).

Table 24: Exact McNemar significance tests against the best model

Compared with best model	Best correct, other wrong	Other correct, best wrong	p-value	Significant
Claude Opus 5.5 (zero-shot)	47	35	0.2242	No
LLM Dhenu2-In-Llama3.2-3B-Instruct (3-shot)	46	25	0.017	Yes
SetFit (all-mpnet-base-v2)	13	14	1.0	No
TF-IDF logistic regression	33	11	0.0013	Yes
Fine-tuned deberta-v3-base	28	7	0.0005	Yes
LLM Dhenu2-In-Llama3.2-3B-Instruct (zero-shot)	99	22	0.0	Yes
RoBERTa Twitter (zero-shot)	101	21	0.0	Yes
LLM Qwen2.5-3B-Instruct (3-shot)	109	17	0.0	Yes
LLM Qwen2.5-3B-Instruct (zero-shot)	156	26	0.0	Yes
VADER + agri lexicon	162	16	0.0	Yes

12.1 Per-Class Performance of the Best Model

Table 25: Per-class performance of the best model

Class	Precision	Recall	F1	Support
Positive	0.873	0.724	0.791	76
Neutral	0.833	0.944	0.885	143
Negative	0.722	0.542	0.619	24

Performance is highest for neutral and positive feedback. Negative feedback is the most difficult class, with only 24 test responses.

12.2 Domain-Specific versus General Language Models

Table 26: Indian-agriculture LLM versus a general LLM of identical size

Model	Setting	Macro-F1
Dhenu2-In-Llama3.2-3B-Instruct (3-shot)	three examples	0.704
Dhenu2-In-Llama3.2-3B-Instruct (zero-shot)	zero-shot	0.445
Qwen2.5-3B-Instruct (3-shot)	three examples	0.391
Qwen2.5-3B-Instruct (zero-shot)	zero-shot	0.321

With three examples, Dhenu2 achieved a macro-F1 of 0.704, compared with 0.391 for Qwen2.5-3B, a general model with the same number of parameters. Agricultural domain knowledge therefore substantially improves the interpretation of farmer feedback.

12.3 Reliability of General-Purpose Tools

VADER and RoBERTa agree only slightly with each other on identical text (Cohen's κ = 0.155), and both perform poorly against human labels (macro-F1 0.223 and 0.439). Sentiment figures produced by these tools without validation should therefore be treated with caution.

Table 27: Agreement between VADER and RoBERTa on the same responses

VADER \ RoBERTa	Negative	Neutral	Positive
Negative	89	104	11
Neutral	29	166	12
Positive	120	483	200

12.4 Effect of Training Data Size

Table 28: Learning curve of the TF-IDF model

Training responses	Macro-F1 (mean)	SD
100	0.494	0.019
200	0.567	0.013
300	0.633	0.021
400	0.641	0.000

Performance increased with every additional 100 labelled responses, indicating that further labelling would yield further gains.

12.5 Aspect-Level Models

Table 29: Aspect-level sentiment evaluation

Method	Aspect mentions detected	Accuracy	Macro-F1	Cohen's κ
Keyword ABSA + VADER (current pipeline)	201 of 274	0.343	0.373	0.191
DeBERTa ABSA (deberta-v3-base-absa-v1.1)	274 of 274	0.522	0.433	0.224

The dedicated DeBERTa aspect model processed every aspect mention and outperformed the keyword method on all metrics.

13. Recommendations (Objective 6)

The following recommendations are addressed to the Department of Agriculture and Farmers Welfare, Government of Andhra Pradesh.

Table 30: Policy recommendations

No.	Recommendation	Supporting evidence
1	Review the enrolment and registration process across all schemes as a single exercise	enrolment ease is the weakest aspect of all four major schemes
2	Make disbursement reliability an operational priority	payment timing accounts for about 44% of complaint text
3	Monitor awareness-to-enrolment conversion for every scheme	conversion ranges from 89.4% to 2.2%
4	Commission a dedicated review of Crop Insurance	lowest conversion among major schemes (9.1%) and the highest negative share
5	Review the adequacy of PM-KISAN's benefit amount	high satisfaction but the lowest self-reported farming impact
6	Add open-ended questions on enrolment and payment timing to monitoring surveys	ratings and written feedback disagree most on these aspects
7	Target additional support at lower-income and marginal farmers	their feedback is consistently the most negative
8	Validate any automated sentiment tool against human labels before use	VADER achieved a macro-F1 of only 0.223

14. Challenges and Solutions

Table 31: Challenges encountered and solutions implemented

Challenge	Solution
one scheme recorded under 93 different spellings	fuzzy matching against a list of canonical scheme names, with every match recorded
answers such as "No problems" scored as negative by lexicon tools	non-informative answers treated as empty before scoring
interviews in Telugu recorded in English	English models used, and the limitation documented
few labelled responses available for training	few-label learning (SetFit), uncertainty-based selection of training responses and AI-suggested, human-verified labels
large language models exceeding laptop memory	4-bit quantisation on the GPU
GPU memory exhausted between consecutive models	explicit memory release and cached predictions so that runs can resume
random variation between training runs	five seeds per trainable model, with mean and standard deviation reported
topic models varying between runs	topic results reported as exploratory

15. Conclusion and Future Scope

15.1 Conclusion

This project shows that farmers' written feedback contains evidence that satisfaction ratings do not capture. Although most surveyed farmers know about at least one scheme, awareness converts unevenly into enrolment, and enrolment is the weakest aspect of every major scheme. Ratings and written feedback disagree most on timeliness and enrolment, so satisfaction scores alone understate exactly the problems farmers describe most often. Payment delays dominate complaints, while farmers' main request is a larger benefit amount.

The model comparison shows that general-purpose sentiment tools are unreliable on farmer feedback, whereas models trained on labelled farmer responses and capable large language models perform substantially better. A domain-specific Indian agriculture LLM clearly outperformed a general LLM of the same size.

15.2 Limitations

- The data cover a single district and a single survey round, so the findings are correlational and hypothesis-generating.
- Interviews were conducted in Telugu and recorded in English, which may reduce nuance.
- Labels were produced jointly as a single consensus set, so inter-annotator agreement could not be measured.
- Negative feedback was under-represented in the test set (24 responses).
- Small schemes, including Crop Insurance (n = 31), require confirmation with larger samples.

15.3 Future Scope

Research publication: The findings are currently being written as a research paper **targeting a Q1 journal**, under the guidance of Prof. Sangeetha S. The principal contributions are the measured gap between farmer ratings and written feedback, the awareness–utilisation gap, and a validated comparison of NLP models including an Indian agriculture LLM.

- Recomputation of the aspect-level findings using the best-performing models.
- Expansion of the labelled dataset with independent double annotation.
- Evaluation of models operating directly on Telugu text.
- Extension to additional districts and survey rounds.
- An interactive dashboard for the Department of Agriculture.

16. Project Links

Table 32: Project links

Resource	Link
Source code and results (GitHub)	https://github.com/singhkrish2006/farmer-scheme-policy-intelligence
Final project presentation (Google Drive)	https://drive.google.com/drive/folders/1Ea17IGBCi2gmo6G_VSptUSgVGAnb_yEK?usp=drive_link
Project report (PDF)	https://github.com/singhkrish2006/farmer-scheme-policy-intelligence/blob/main/docs/Complete_Project_Report.pdf

The raw survey data and labelling worksheets are not published, as they contain personal information of the participants.

Appendix

A. Execution Steps

Table 33: Execution steps

Step	Command	Function
1	pip install -r requirements.txt	installation of dependencies
2	python scripts/01_run_pipeline.py	cleaning and initial sentiment models
3	python scripts/02_run_analysis.py	statistics and topic models
4	python scripts/05_evaluate_models.py	evaluation against human labels
5	python scripts/12_compare_models.py	comparison of eleven models

B. LLM Prompt

> You label farmers' feedback about Indian government welfare schemes. Reply with exactly one word: Positive, Negative or Neutral.
> Positive = mainly expresses satisfaction, benefit, relief or approval.
> Negative = mainly expresses a problem, delay, frustration or complaint.
> Neutral = factual with no clear tone, or genuinely mixed with no dominant tone.
>
> Few-shot: 3 training examples per class, seed 42; greedy decoding

C. Image Credits

Table 34: Image credits

Image	Author	Licence	Source
Farmer ploughing with cattle, Nandyal District, Andhra Pradesh.jpg	PJeganathan	CC BY-SA 3.0	Wikimedia Commons
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